

STUDENT PROJECT · TEAM NO. 52

# FLIGHT PRICE PREDICTION USING LINEAR REGRESSION


DOCUMENTATION

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PRESENTATION

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EXECUTION

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VIDEO

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Project link: github.com/adityapatil1807/Flight-Price-Predictions-

02 · PROBLEM FORMULATION

# THE PROBLEM AND DATASET

### Goal

Predict a flight ticket price from route, timing, class, and booking lead time.

### Dataset source

Kaggle, with flight-ticket data between six Indian cities.

### Target

price



Six Indian cities · flight-ticket data · price prediction

MAIN COLUMNS

airline · flight · source_city · departure_time · stops · arrival_time · destination_city · class · duration · days_left · price

SELECTED INPUTS

source_city · destination_city · departure_time · class · days_left

# MODEL ARCHITECTURE & TECHNOLOGIES USED



Input data becomes a repeatable prediction pipeline.

1

## Read

Read and understand the dataset.

2

## Explore

Explore price patterns with a histogram, a days-left scatter plot, and average-price comparisons by departure time.

3

## Prepare

Prepare the data by removing duplicates, handling missing values, and one-hot encoding categorical features.

4

## Split and train

Split the data into 80% training and 20% testing, then train a Linear Regression model.

5

## Evaluate

Evaluate with MAE, MSE, RMSE, and $R^2$, then predict a user flight price from route, time, class, and booking days.

LIBRARIES

pandas · NumPy · Matplotlib · scikit-learn

04 · TEST SET EVALUATION

# MODEL PERFORMANCE & EVALUATION

Demo Screenshot and Keys Number

MAE

**4,673.923**

MSE

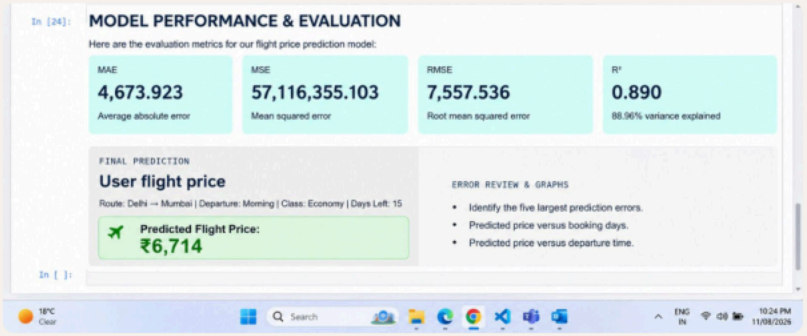
**57,116,355.103**

RMSE

**7,557.536**

$R^2$

**0.890**



# PROJECT LEARNINGS & TEAM CONTRIBUTIONS

## WHAT WE LEARNED

- Categorical flight details must be converted into model-ready values.
- The model should be tested on data it did not train on.

## WHAT WORKED

One preprocessing-and-regression pipeline made prediction organized and repeatable.

## WHAT DIDN'T WORK AS EXPECTED

The submitted notebook has no executed evaluation output. MAE, MSE, RMSE, $R^2$, and the final prediction must be recorded after running it.

## TEAM CONTRIBUTIONS

| Role | Contributor |
|---|---|
| Documentation | Aditya Patil |
| Presentation | Nipun Singhal |
| Execution lead | Shaurya Kamboj |
| Video | Jarmik |

**AI disclosure:** ChatGPT and Claude supported explanations, documentation, and presentation wording; they did not train the model or produce **project results.**